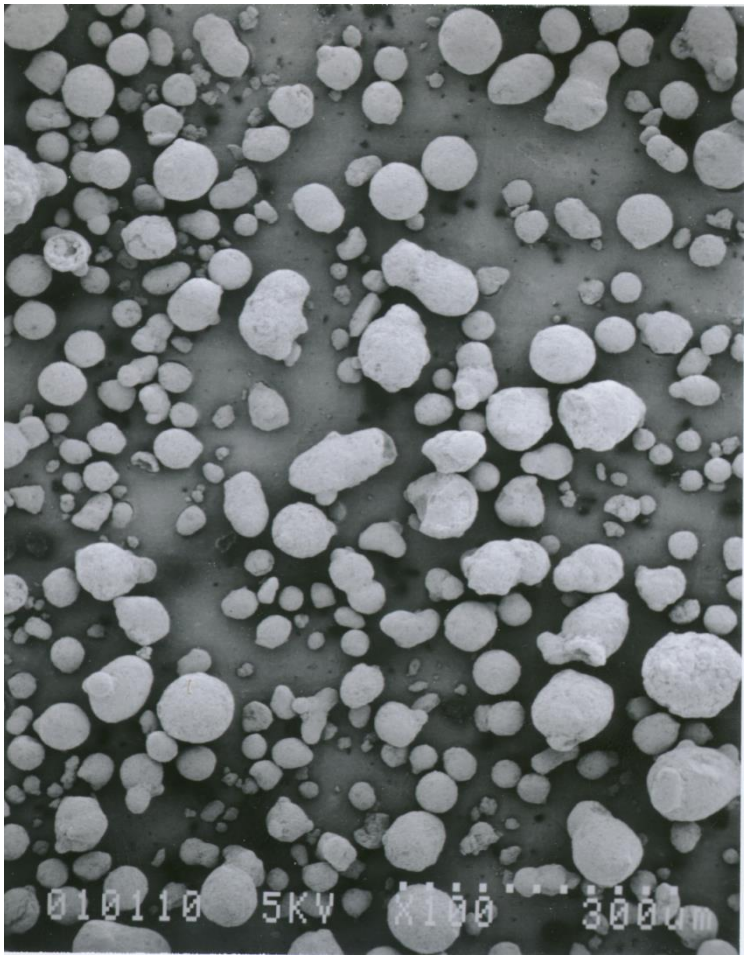
Catalytic Reactions and Reactors

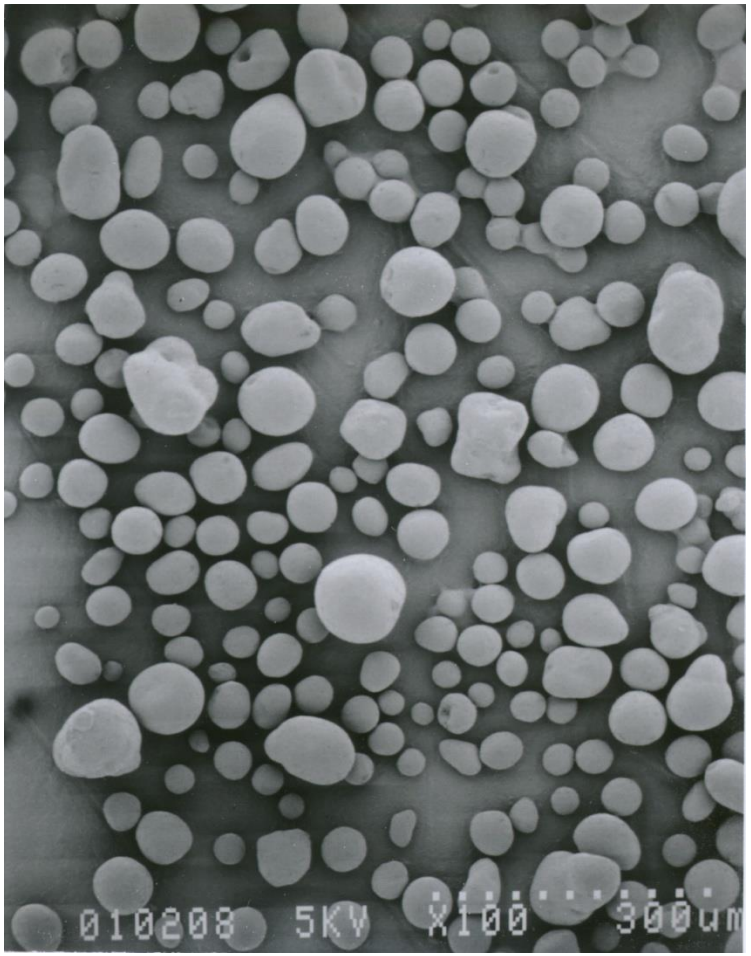
CENG252
Zheng Chen
(zhc199@ucsd.edu)
10/22/2026



Silica-Alumina Cat-Cracking Catalyst (100X)

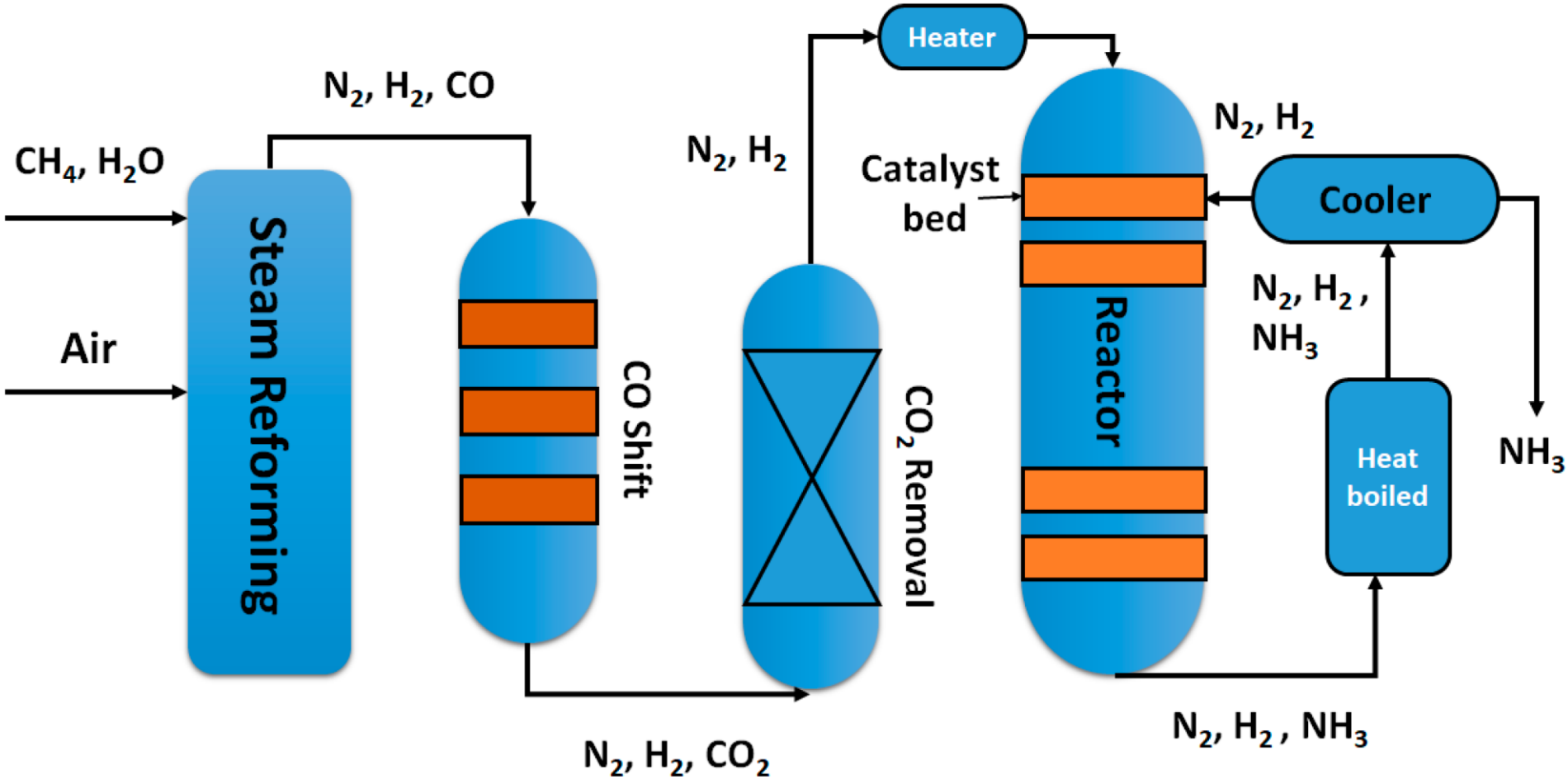


fresh



spent

A Flow Scheme for the Haber-Bosch Process



A Mechanism that Involves the Heterogeneous-Catalyst

$$N_{2(g)} \rightarrow N_{2(adsorbed)} \quad (1)$$

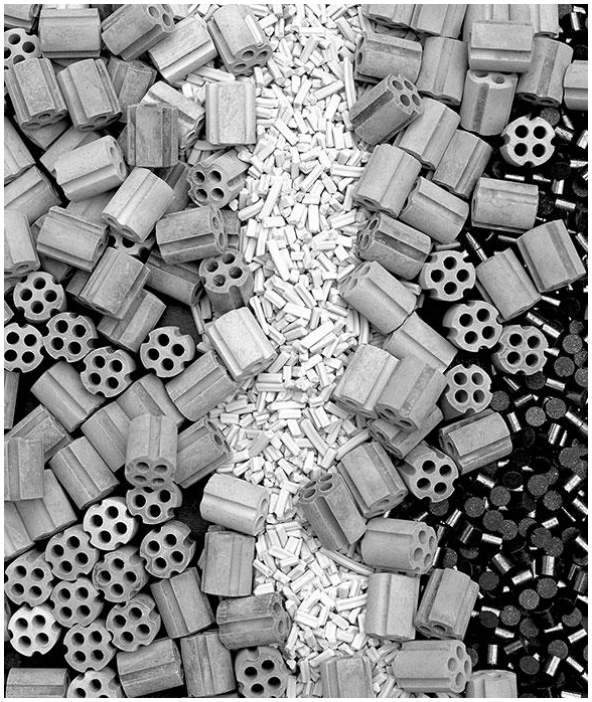
$$N_{2(adsorbed)} \rightarrow 2N_{(adsorbed)} \quad (2)$$

$$H_{2(g)} \rightarrow H_{2(adsorbed)} \quad (3)$$

$$H_{2(adsorbed)} \rightarrow 2H_{(adsorbed)} \quad (4)$$

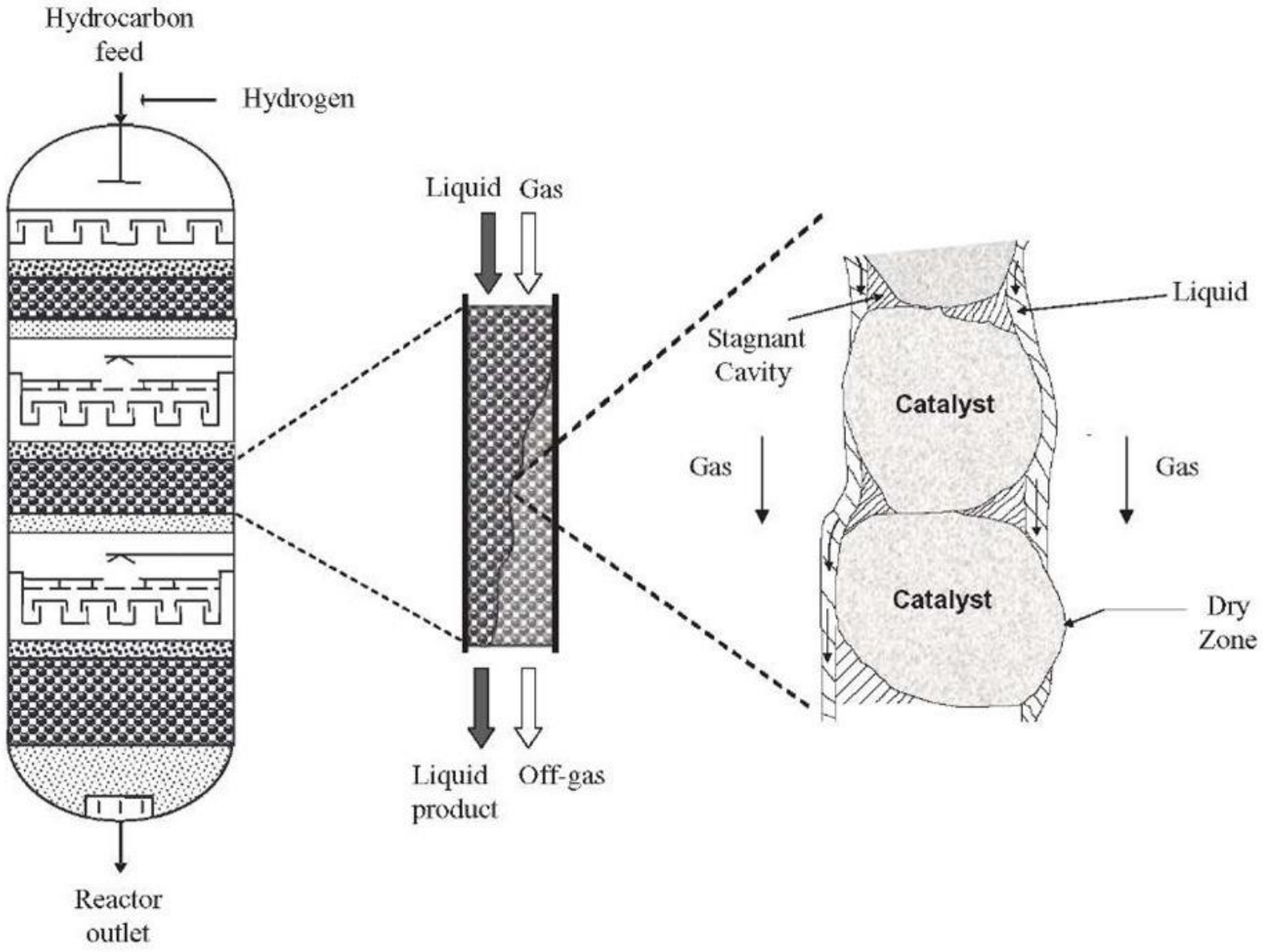
$$N_{(adsorbed)} + 3H_{(adsorbed)} \rightarrow NH_{3(adsorbed)} \quad (5)$$

$$NH_{3(adsorbed)} \rightarrow NH_{3(g)} \quad (6)$$



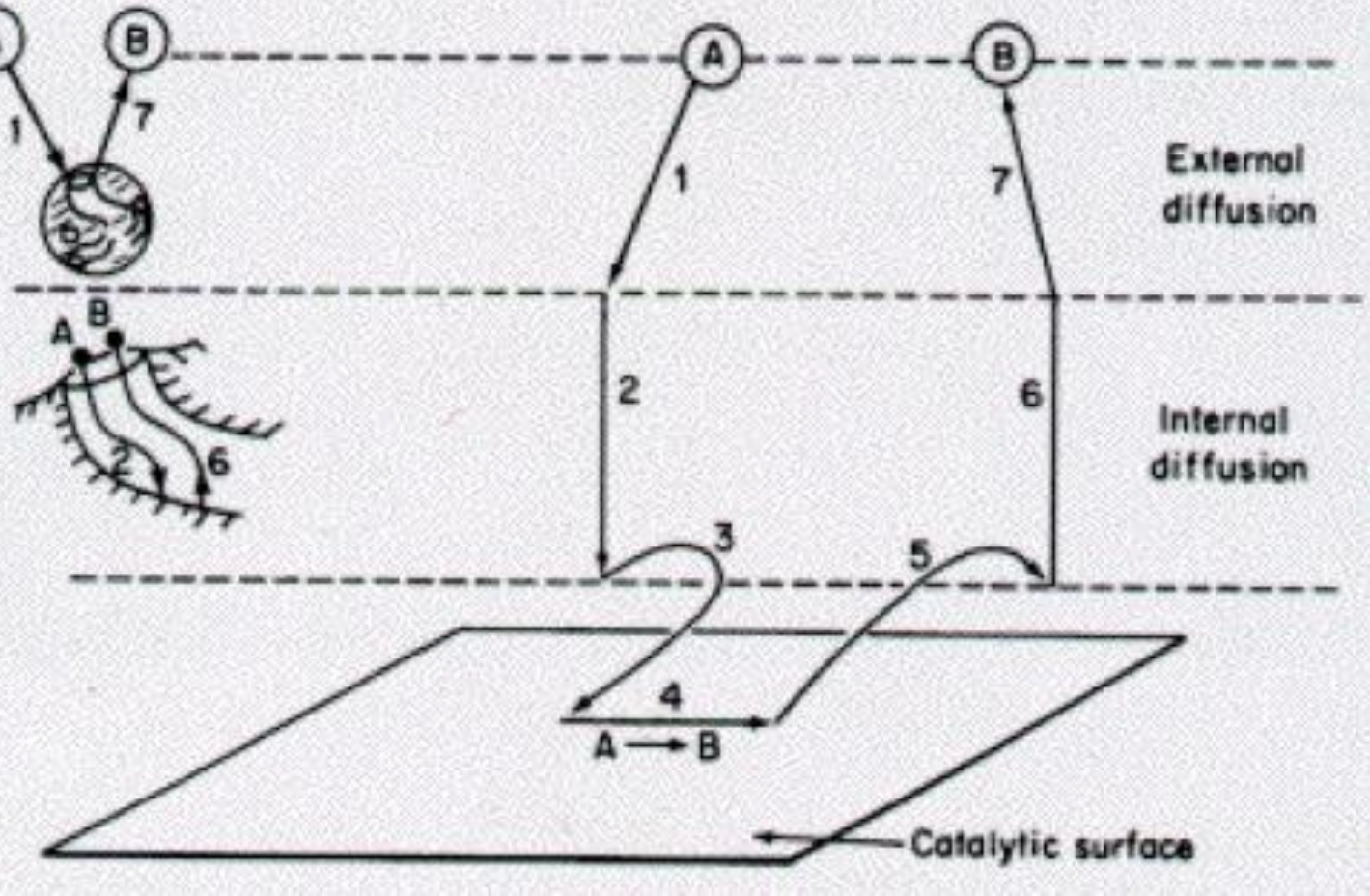
The most common catalyst: K_2O or Al_2O_3-promoted iron catalyst

Catalytic Reactor (hydrocarbon reforming)



Steps in Heterogeneous Catalytic Reaction

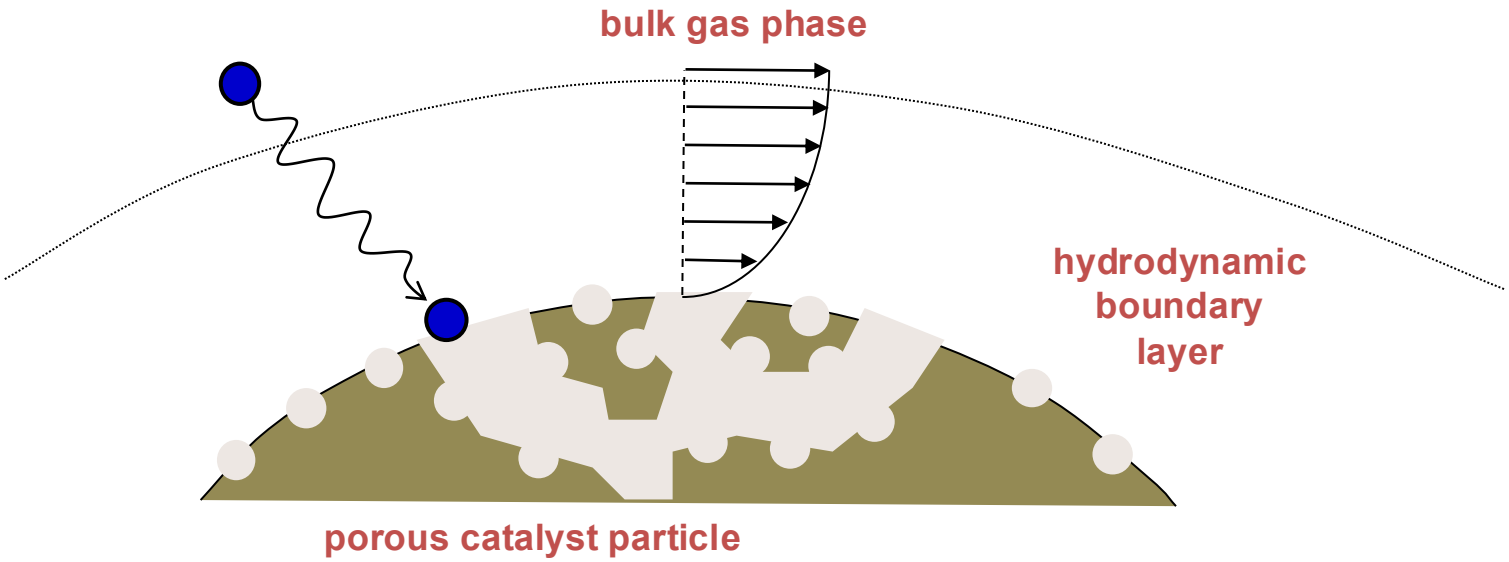
A reaction takes place *on* the surface, but the species involved in the reaction must get *to* and *from* the surface



Page 656

Transport & Kinetic Processes in Catalytic Reactions

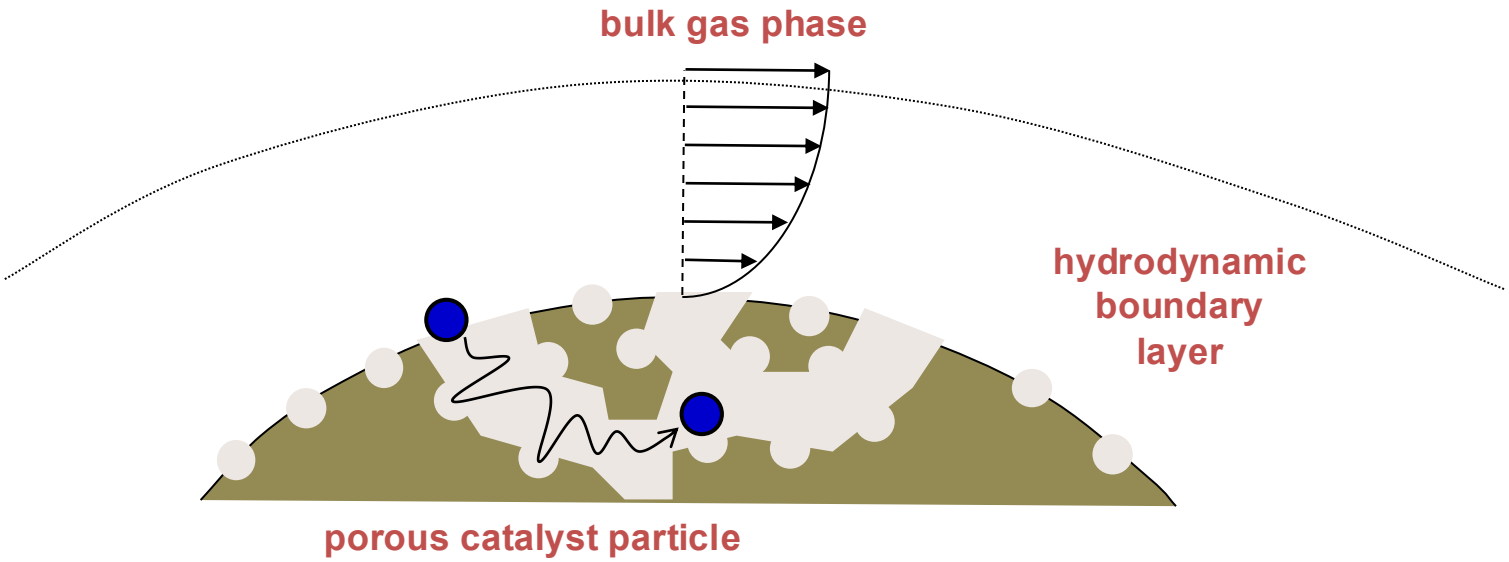
external diffusion of A



Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

internal diffusion of A



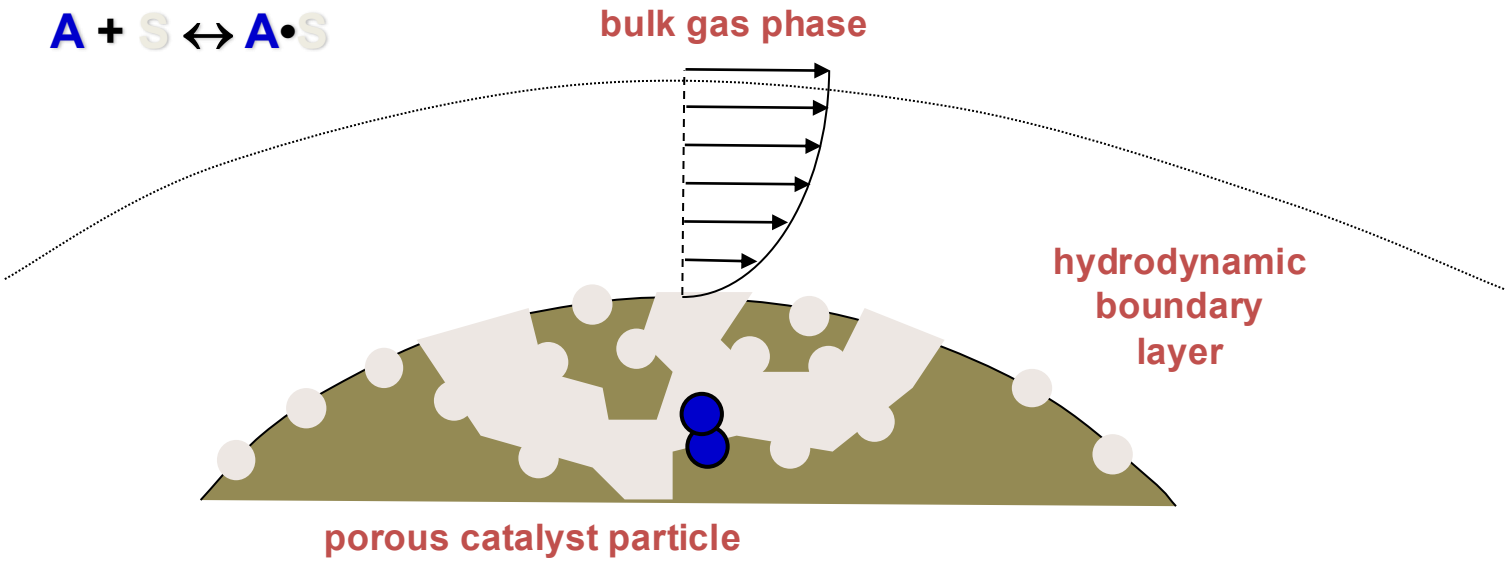
Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

internal diffusion of A

adsorption of A

$A + S \leftrightarrow A \bullet S$



S ≡ catalyst adsorption site

Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

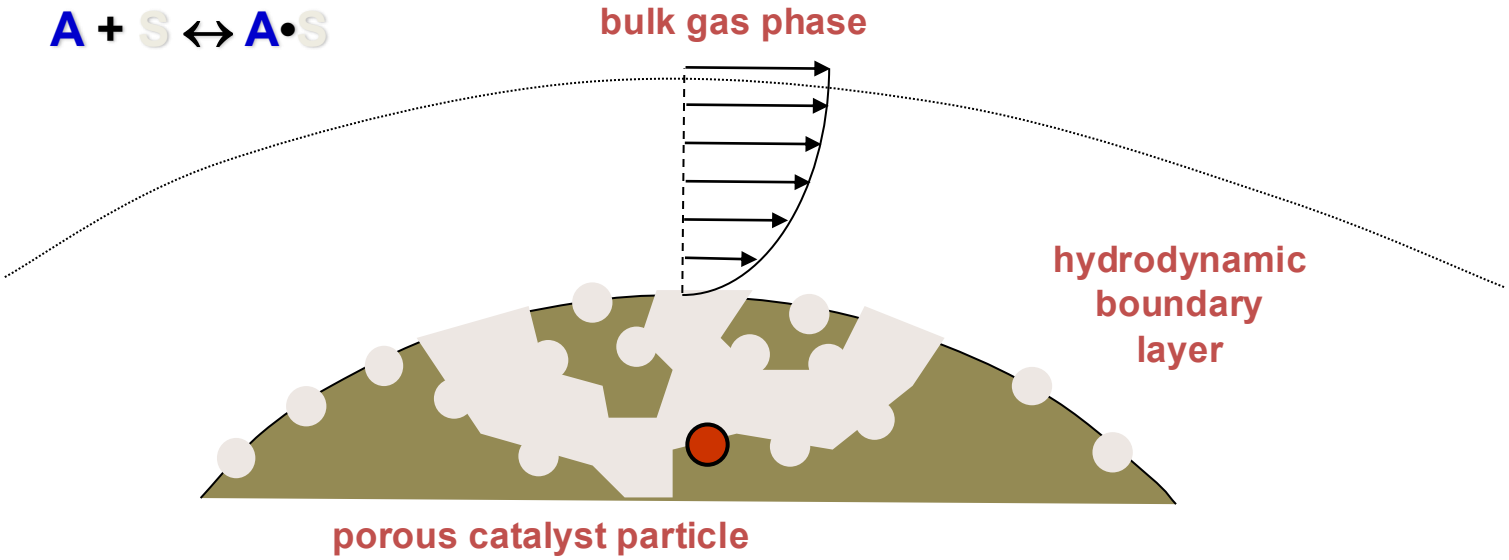
internal diffusion of A

adsorption of A

$A + S \leftrightarrow A{\bullet}S$

reaction of A to B

$A{\bullet}S \rightarrow B{\bullet}S$



Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

internal diffusion of A

adsorption of A

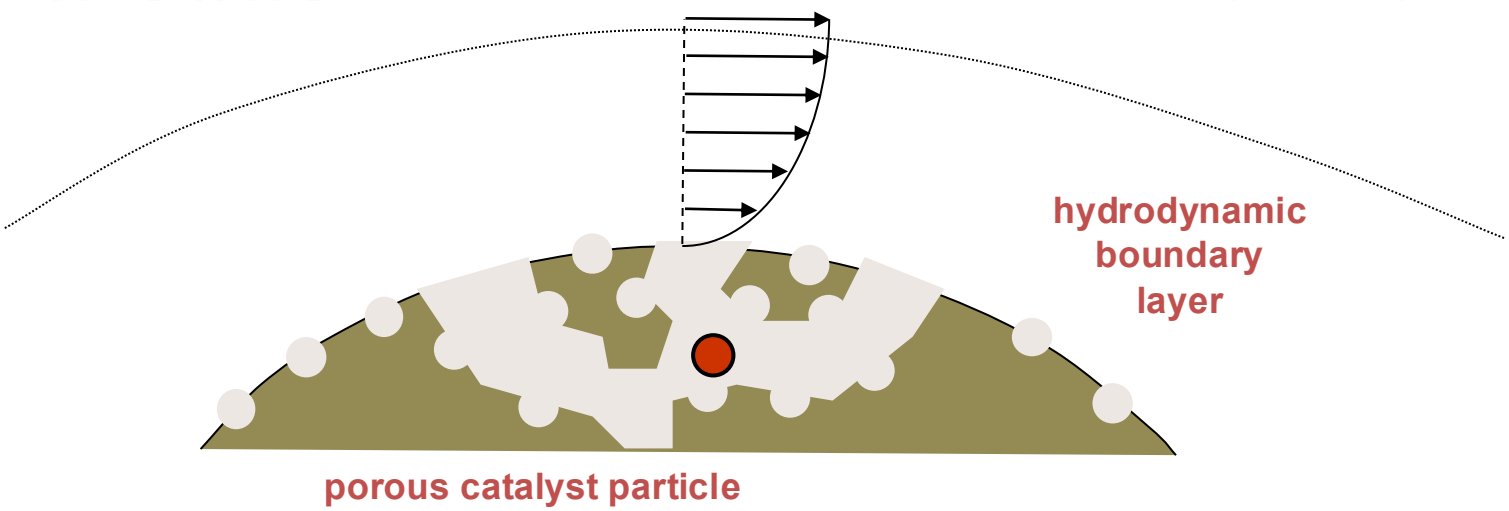
$A + S \leftrightarrow A{\bullet}S$

reaction of A to B

$A{\bullet}S \rightarrow B{\bullet}S$

desorption of B

$B{\bullet}S \leftrightarrow B + S$



Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

internal diffusion of A

adsorption of A

$A + S \leftrightarrow A{\bullet}S$

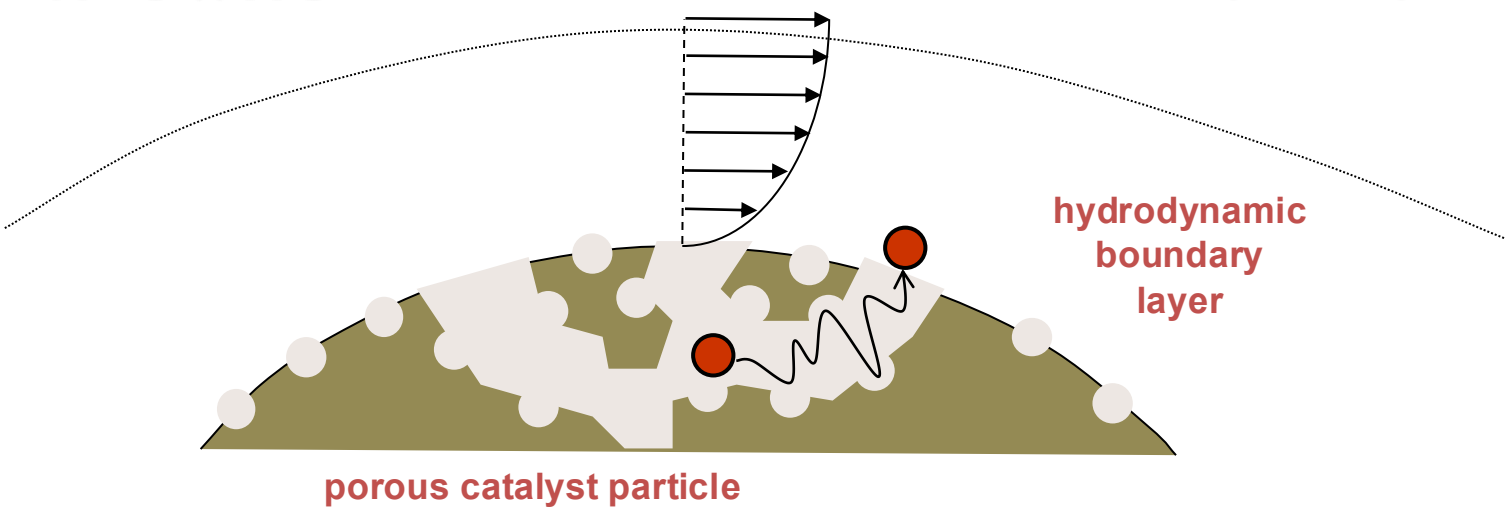
reaction of A to B

$A{\bullet}S \rightarrow B{\bullet}S$

internal diffusion of B

desorption of B

$B{\bullet}S \leftrightarrow B + S$



Transport & Kinetic Processes in Catalytic Reactions

external diffusion of A

internal diffusion of A

adsorption of A

$A + S \leftrightarrow A{\bullet}S$

reaction of A to B

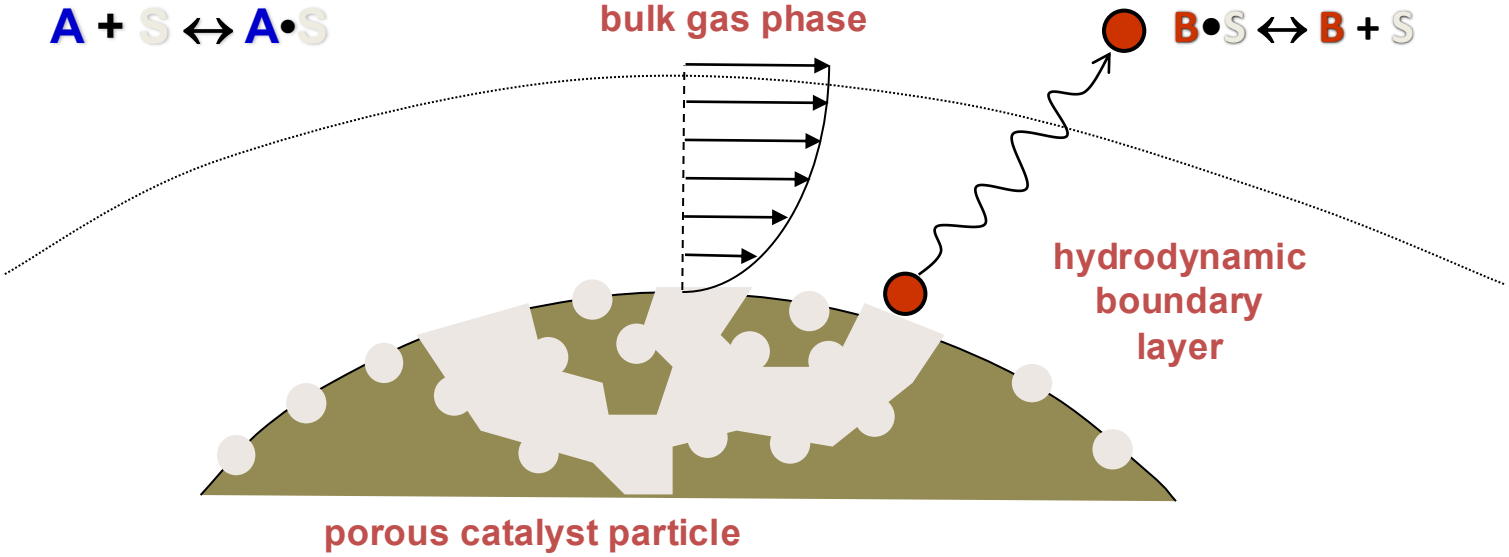
$A{\bullet}S \rightarrow B{\bullet}S$

external diffusion of B

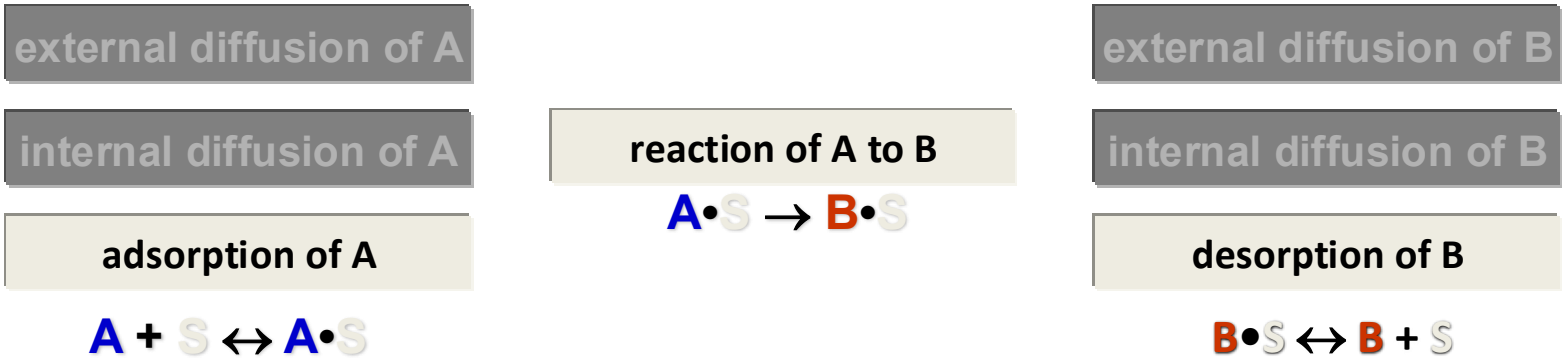
internal diffusion of B

desorption of B

$B{\bullet}S \leftrightarrow B + S$



Transport & Kinetic Processes in Catalytic Reactions



- One of these seven transport and kinetic processes occurs the slowest. We say that step is "rate-limiting"
- It is necessary to determine the "rate-limiting" step to analyze the kinetics.
- Start by restricting ourselves to these steps…

Molecular Adsorption

adsorption of A

$A + S \leftrightarrow A{\bullet}S$ **Treat as an elementary reaction**

site balance: $C_t = C_v + C_{A\bullet S}$

rate of adsorption: $r_{ad_A} = k_A P_A C_v$

rate of desorption: $r_{de_A} = k_{-A} C_{A\bullet S}$

net rate of sorption: at equilibrium $r_{a/d_A} = k_A \left(P_A C_v - \frac{C_{A\bullet S}}{K_A} \right) = 0$

$r_A \propto P_A$

pressure is a measure of collision frequency (from the molecular theory of gases)

$$C_{A\bullet S} = \frac{K_A P_A C_t}{1 + K_A P_A}$$

$K_A \equiv$ adsorption equilibrium constant (k_A/k_{-A})

$C_t \equiv$ total concentration of sites per mass catalyst

$C_v \equiv$ concentration of vacant sites per mass catalyst

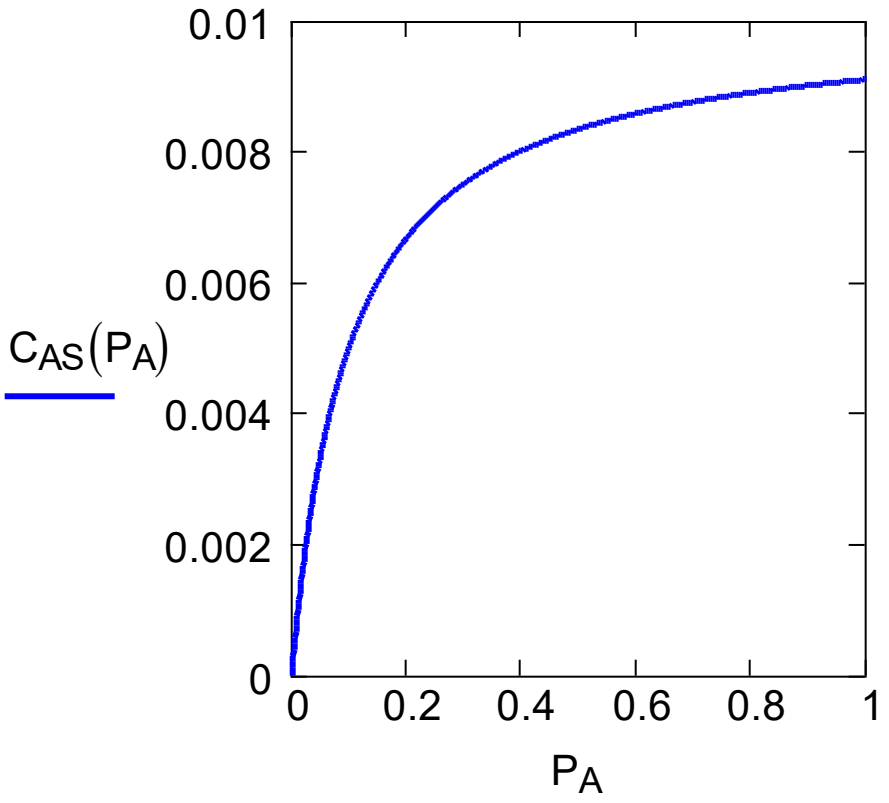
$C_{A\cdot S} \equiv$ surface concentration of sites occupied by species A, mol/g cat

Langmuir Adsorption Isotherm (P664)

adsorption of A

$$A + S \leftrightarrow A{\bullet}S$$

note that we've treated both adsorption and desorption in this analysis



Assumes

- **monolayer coverage of surface**
- **uniform surface**
- **Interaction between gas/site only**

Langmuir Isotherm describes the equilibrium partition of gas between absorbed and desorbed states.

$$C_{A\bullet S} = \frac{K_A P_A C_t}{1 + K_A P_A}$$

Langmuir Adsorption Isotherm

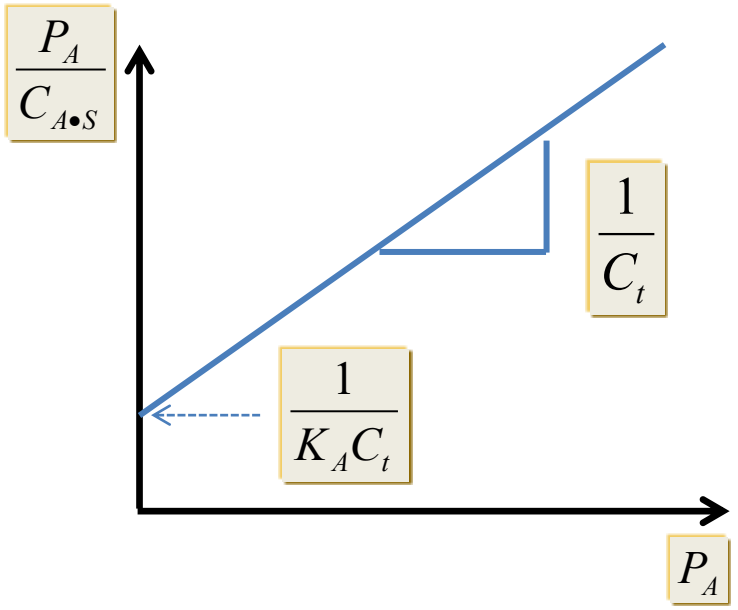
$$\frac{P_A}{C_{A\bullet S}} = \frac{1}{K_A C_t} + \frac{P_A}{C_t}$$

Molecular Adsorption

$$C_{A\bullet S} = \frac{K_A P_A C_t}{1 + K_A P_A}$$

Langmuir Adsorption Isotherm

$$\frac{P_A}{C_{A\bullet S}} = \frac{1}{K_A C_t} + \frac{P_A}{C_t}$$



The linear plot will determine if the data Conform to a Langmuir single-site isotherm.

Atomic (Dissociative) Adsorption (P665-666)

adsorption of A_2

$$A_2 + 2S \leftrightarrow 2A\bullet S$$

site balance:

$$C_t = C_v + C_{A\bullet S}$$

rate of adsorption:

$$r_{ad_A} = k_A P_A C_v^2$$

$$C_{A\bullet S} = \frac{(K_A P_A)^{1/2} C_t}{1 + (K_A P_A)^{1/2}}$$

rate of desorption:

$$r_{de_A} = k_{-A} C_{A\bullet S}^2$$

net rate of sorption (at equilibrium)

$$r_{a/d_A} = k_A \left(P_A C_v^2 - \frac{C_{A\bullet S}^2}{K_A} \right) = 0$$

$K_A = k_A / k_{-A}$

Multicomponent Adsorption (P666)

adsorption of A and B

$$A + S \leftrightarrow A{\bullet}S \qquad B + S \leftrightarrow B{\bullet}S$$

site balance:

$$C_t = C_v + C_{A\bullet S} + C_{B\bullet S}$$

rate of adsorption:

$$r_{ad_A} = k_A P_A C_v$$

rate of desorption:

$$r_{de_A} = k_{-A} C_{A\bullet S}$$

$$C_{A\bullet S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B}$$

net rate of sorption (at equilibrium)

$$r_{a/d_A} = k_A \left(P_A C_v - \frac{C_{A\bullet S}}{K_A} \right) = 0$$

get $C_{A\bullet S}$

Surface Reaction (P666-669)

single site mechanism

$$A\bullet S \leftrightarrow B\bullet S$$

(e.g. isomerization)

Langmuir-Hinshelwood

dual site mechanisms

$$A\bullet S + S \leftrightarrow B\bullet S + S$$

(e.g. dehydration)

adjacent-site interaction

$$A\bullet S + B\bullet S \leftrightarrow C\bullet S + D\bullet S$$

(e.g. $CO+O \rightarrow CO_2$)

adjacent-species interaction

$$A\bullet S + B\bullet S' \leftrightarrow C\bullet S' + D\bullet S$$

different site types

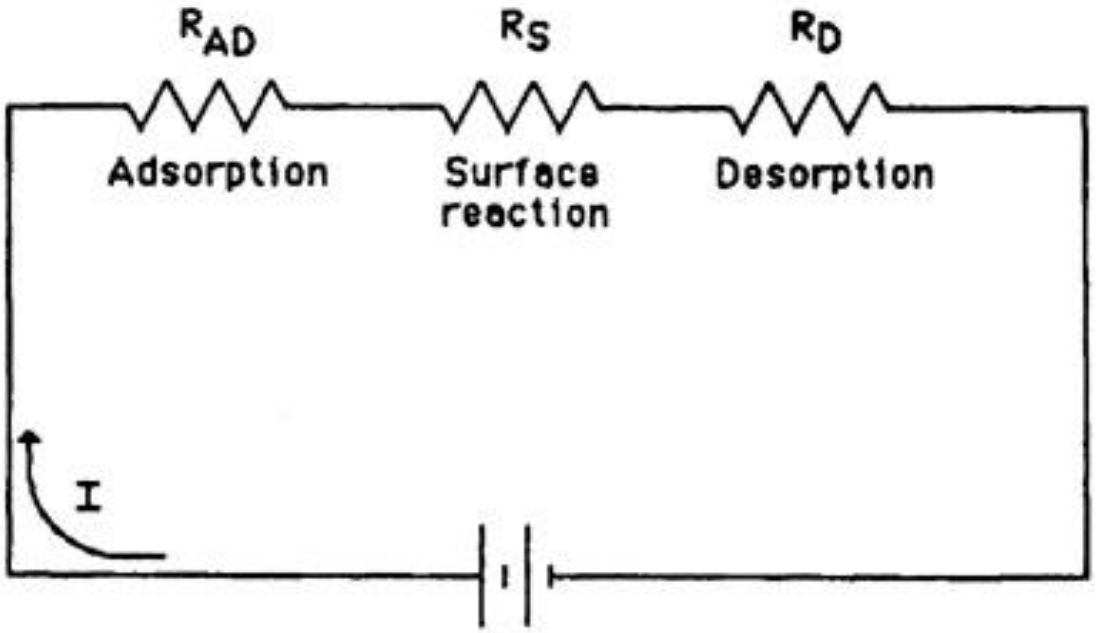
gas phase interaction

$$A\bullet S + B_{(g)} \leftrightarrow C\bullet S$$

Eley-Rideal Kinetics

The Rate Limiting Step

Which step has the largest resistance?



Electrical analog to heterogeneous reactions

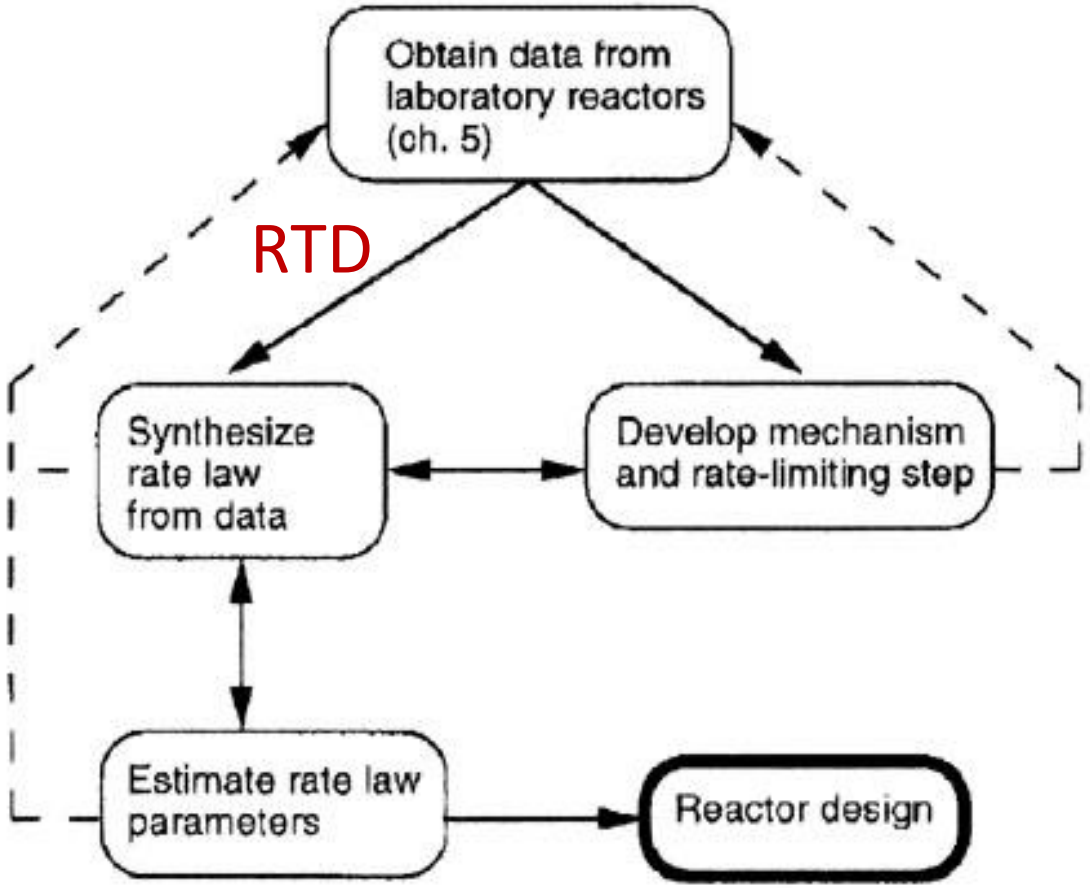
Rate Limiting Step (P669)

- Steady state, heterogeneous reactions;

$$-r_A = r_{a_A} = r_{s_A} = r_{d_A}$$

- Solution Algorithm:
  - Select a mechanism (as molecular or atomic adsorption; single or dual-site reaction: ***L-H mechanism***)
  - Assume a rate limiting step
  - Find the expression for $C_{i\bullet S}$ using the steps that are not rate limiting (occur at equilibrium)
  - Write the site balance
  - Derive the rate law
  - Compare with experimental data

Collecting and Analyzing Data



Collecting information for catalytic reactor design

Example 1: Surface Reaction Rate-Limiting (> 75% of all rxns)

Select a mechanism

$$A + S \leftrightarrow A\bullet S$$

Assume a rate limiting step →

$$A\bullet S \leftrightarrow B\bullet S$$

$$B\bullet S \leftrightarrow B + S$$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at high rate/fast)

$$\frac{r_{a/d_A}}{k_A} = \left(P_A C_v - \frac{C_{A\bullet S}}{K_A} \right) \cong 0 \qquad C_{A\bullet S} = K_A P_A C_v$$

$$\frac{r_{a/d_B}}{k_{-B}} = \left(C_{B\bullet S} - K_B P_B C_v \right) \cong 0 \qquad C_{B\bullet S} = K_B P_B C_v$$

Example 1

Select a mechanism $A + S \leftrightarrow A\bullet S$

Assume a rate limiting step → $A\bullet S \leftrightarrow B\bullet S$

$B\bullet S \leftrightarrow B + S$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at high rate/fast)

Write the site balance $C_t = C_v + C_{A\bullet S} + C_{B\bullet S}$

Derive the rate law

$$C_{A\bullet S} = K_A P_A C_v$$

$$r_s = k_s \left(C_{A\bullet S} - \frac{C_{B\bullet S}}{K_s} \right) \neq 0$$

$$C_{B\bullet S} = K_B P_B C_v$$

Example 1

$$C_t = C_v + K_A P_A C_v + K_B P_B C_v$$

$$C_t = C_v\left(1 + K_A P_A + K_B P_B\right)$$

$$C_v = C_t\left(1 + K_A P_A + K_B P_B\right)^{-1}$$

$$r_s = k_s C_v\left(K_A P_A - \frac{K_B}{K_s} P_B\right)$$

$$A + S \leftrightarrow A{\bullet}S$$

rate limiting → $A{\bullet}S \leftrightarrow B{\bullet}S$

$$B{\bullet}S \leftrightarrow B + S$$

$$r_s = \frac{k_s K_A C_t\left(P_A - K_e P_B\right)}{\left(1 + K_A P_A + K_B P_B\right)} = -r_A'$$

(k = $k_s K_A C_t$)

$$K_e = \frac{K_B}{K_A K_s}$$

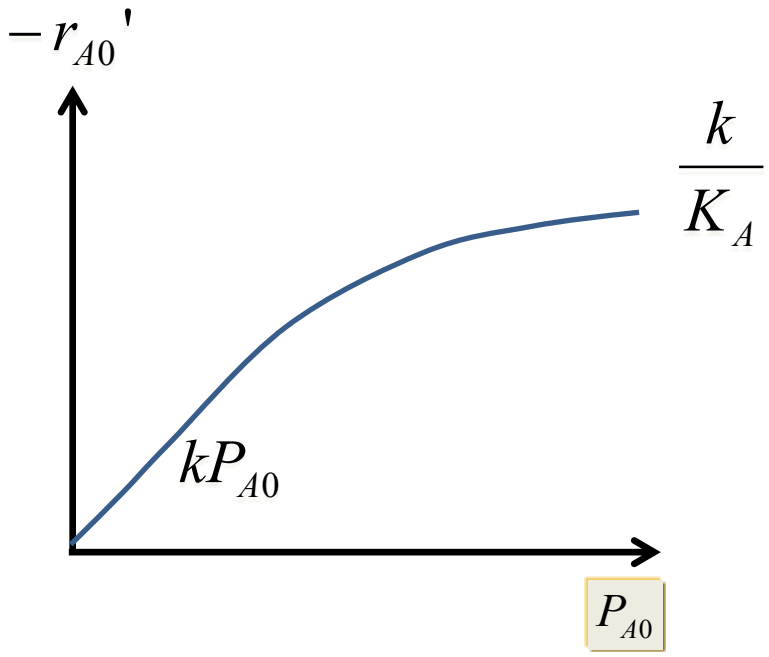
Example 1: The Initial Rate

$$r_s = \frac{\overbrace{k_s K_A C_t}^{k}\left(P_A - K_e P_B\right)}{\left(1 + K_A P_A + K_B P_B\right)} = -r_A'$$



$$-r_{A0}' = \frac{kP_{A0}}{1 + K_A P_{A0}}$$

(small P_{B0})



Example 2: Adsorption Rate-Limiting

another rate limiting step → $A + S \leftrightarrow A{\bullet}S$

$A{\bullet}S \leftrightarrow B{\bullet}S$

$B{\bullet}S \leftrightarrow B + S$

Same mechanism (again)

Find $C_{i\text{-}S}$ (steps not rate-limiting occur at equilibrium)

$$\frac{r_s}{k_s} = \left(C_{A\bullet S} - \frac{C_{B\bullet S}}{K_s} \right) \cong 0 \qquad C_{A\bullet S} = \frac{C_{B\bullet S}}{K_s}$$

$$\frac{r_{a/d_B}}{k_{-B}} = \left(C_{B\bullet S} - K_B P_B C_v \right) \cong 0 \qquad C_{B\bullet S} = K_B P_B C_v$$

Example 2

another rate limiting step → $A + S \leftrightarrow A\bullet S$

Same mechanism (again)

$A\bullet S \leftrightarrow B\bullet S$

$B\bullet S \leftrightarrow B + S$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at equilibrium)

Write the site balance $C_t = C_v + C_{A\bullet S} + C_{B\bullet S}$

Derive the rate law

$$r_{a/d_A} = k_A\left(P_A C_v - \frac{C_{A\bullet S}}{K_A}\right) \neq 0$$

$$C_{A\bullet S} = \frac{C_{B\bullet S}}{K_s}$$

$$C_{B\bullet S} = K_B P_B C_v$$

Example 2

$$C_t = C_v + \frac{K_B P_B C_v}{K_s} + K_B P_B C_v$$

rate limiting →

$$A + S \leftrightarrow A{\bullet}S$$

$$A{\bullet}S \leftrightarrow B{\bullet}S$$

$$B{\bullet}S \leftrightarrow B + S$$

$$C_t = C_v\left(1 + \frac{K_B P_B}{K_s} + K_B P_B\right)$$

$$C_v = C_t\left[1 + \left(K_B + K_e K_A\right)P_B\right]^{-1}$$

$$r_s = \frac{k_A C_t\left(P_A - \frac{P_B}{K_e}\right)}{1 + P_B\left(K_B + K_e K_B\right)}$$

$$r_{a/d_A} = k_A C_v\left(P_A - \frac{K_B}{K_s K_A} P_B\right)$$

$$K_e = \frac{K_A K_s}{K_B}$$

Example 2: The Initial Rate

(small P_{B0})

$$r_s = \frac{k_A C_t \left(P_A - \frac{P_B}{K_e} \right)}{1 + P_B \left(K_B + K_e K_B \right)}$$

